

JKBOSE 2026

CLASS 11TH

**INTRODUCTION TO THREE
DIMENSIONAL GEOMETRY**

MATHS

LECTURE-3

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TOPICS TO BE COVERED

1

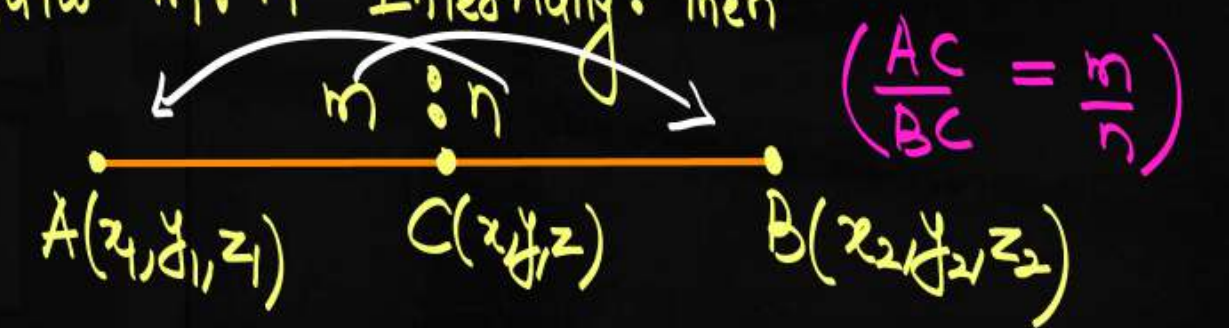
Exercise Discussion 11.2

2

Miscellaneous Exercise

SECTION FORMULA:

★ **Internal Division:** Suppose that a point $C(x, y, z)$ divides the line segment joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in the ratio $m:n$ internally. Then

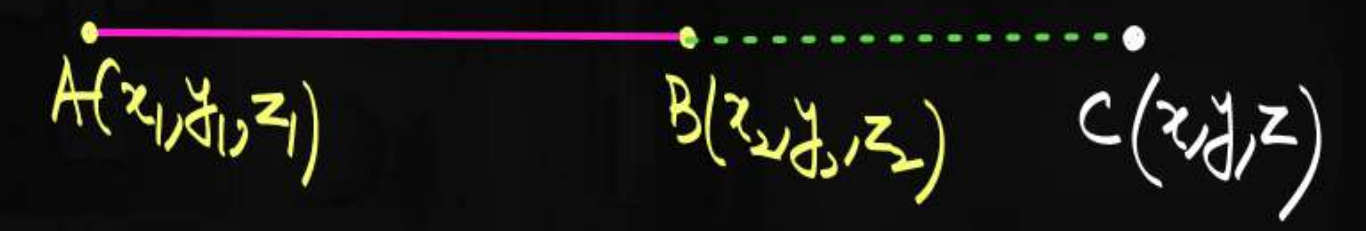


$$(x, y, z) = C\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$$

* If C is the mid-point of AB , then

$$C(x, y, z) = C\left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2}, \frac{z_2 + z_1}{2}\right)$$

★ **External Division:** Suppose a point $C(x, y, z)$ divides the line segment joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in the ratio $m:n$ externally. Then $\frac{AC}{BC} = \frac{m}{n}$



$$C(x, y, z) = C\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n}\right)$$

QUESTION

The ratio in which the line joining $(1, 2, 3)$ and $(-3, 4, -5)$ is divided the XY - plane is:

- 1 $5:3$ ✗
- 2 $3:5$ ✓
- 3 $2:3$ ✗
- 4 None of these ✗

(Sol) Suppose that XY -plane divides the line segment joining $A(1, 2, 3)$ and $B(-3, 4, -5)$ in $k:1$

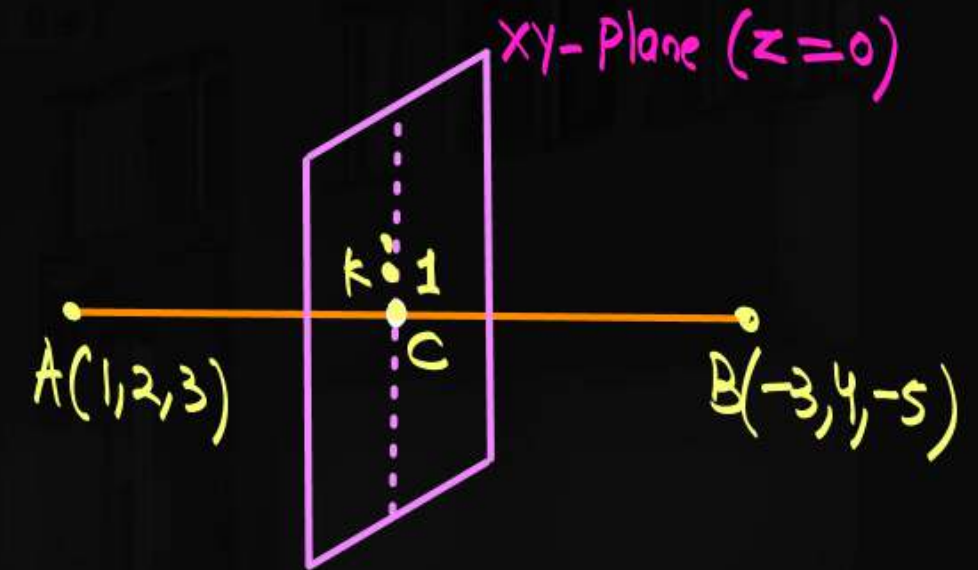
$$\text{Then } C(x, y, z) = C\left(\frac{-3k+1}{k+1}, \frac{4k+2}{k+1}, \frac{-5k+3}{k+1}\right)$$

Since C lies on the XY -plane,
 z -Co-ordinate of $C = 0$

$$\frac{-5k+3}{k+1} = 0$$

$$\Rightarrow -5k+3=0$$

$$\Rightarrow k = \frac{3}{5}$$



So Required ratio $= \frac{3}{5} : 1$
 $= 3:5$

QUESTION

A point R with x-coordinate 4 lies on the line segment joining the points $P(2, -3, 4)$ and $Q(8, 0, 10)$ then the coordinates of the point R is :

- 1 ☒ (4, -2, -6)
- 2 ☒ (4, 2, 6)
- 3 ☒ (4, -2, 6)
- 4 ☒ None of these

(Sol) By section formula

$$R(4, y, z) = R\left(\frac{8k+2}{k+1}, \frac{-3}{k+1}, \frac{10k+4}{k+1}\right)$$

$$\Rightarrow \frac{8k+2}{k+1} = 4$$

$$\Rightarrow 8k+2 = 4k+4$$

$$\Rightarrow 8k-4k = 4-2$$

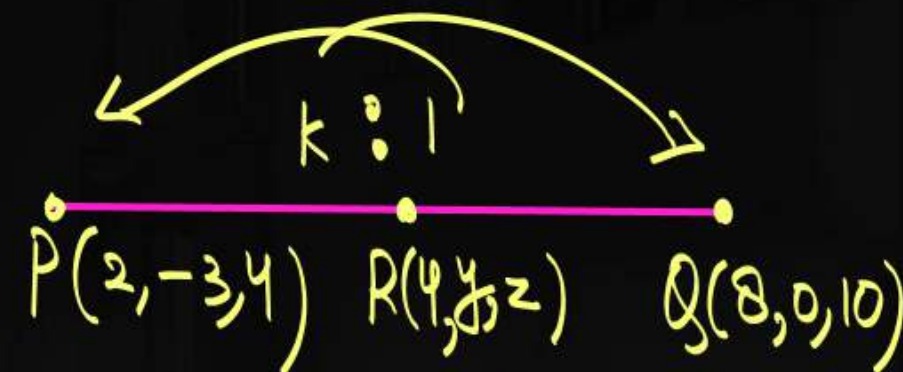
$$\Rightarrow 4k = 2$$

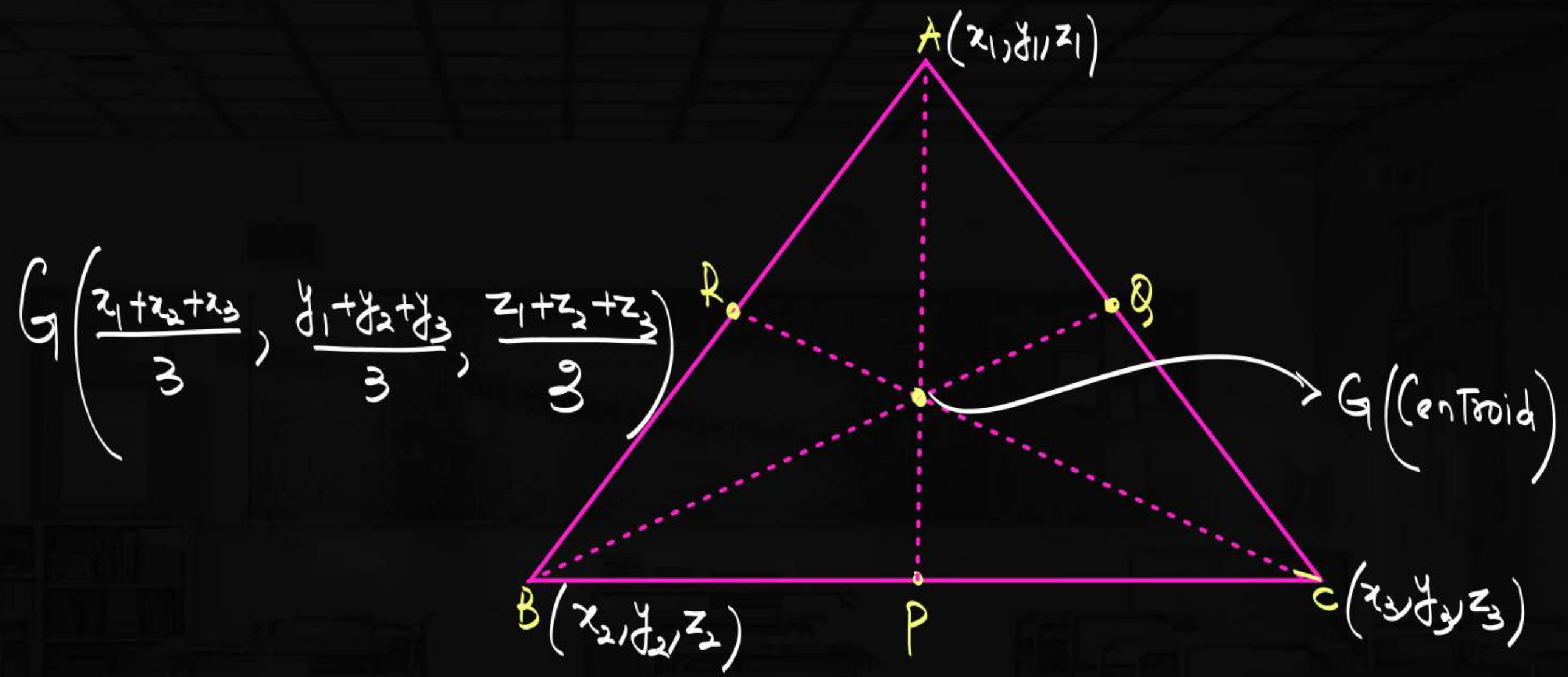
$$\Rightarrow k = \frac{1}{2}$$

So, Required point = $R\left(4, -\frac{3}{\frac{1}{2}+1}, \frac{10(\frac{1}{2})+4}{\frac{1}{2}+1}\right)$

$$= R\left(4, -\frac{3}{\frac{3}{2}}, \frac{5+4}{\frac{3}{2}}\right)$$

$$= R\left(4, -2, 6\right)$$





QUESTION

If the origin is the centroid of ΔPQR with vertices $P(2a, 2, 6)$, $Q(-4, 3b, -10)$ and $R(8, 14, 2c)$, then the values of a , b and c are, respectively

- 1 ☒ $-2, -\frac{16}{3}, 3$
- 2 ☒ $2, \frac{16}{3}, -2$
- 3 ☒ $-2, -\frac{16}{3}, 2$
- 4 ☒ None of these

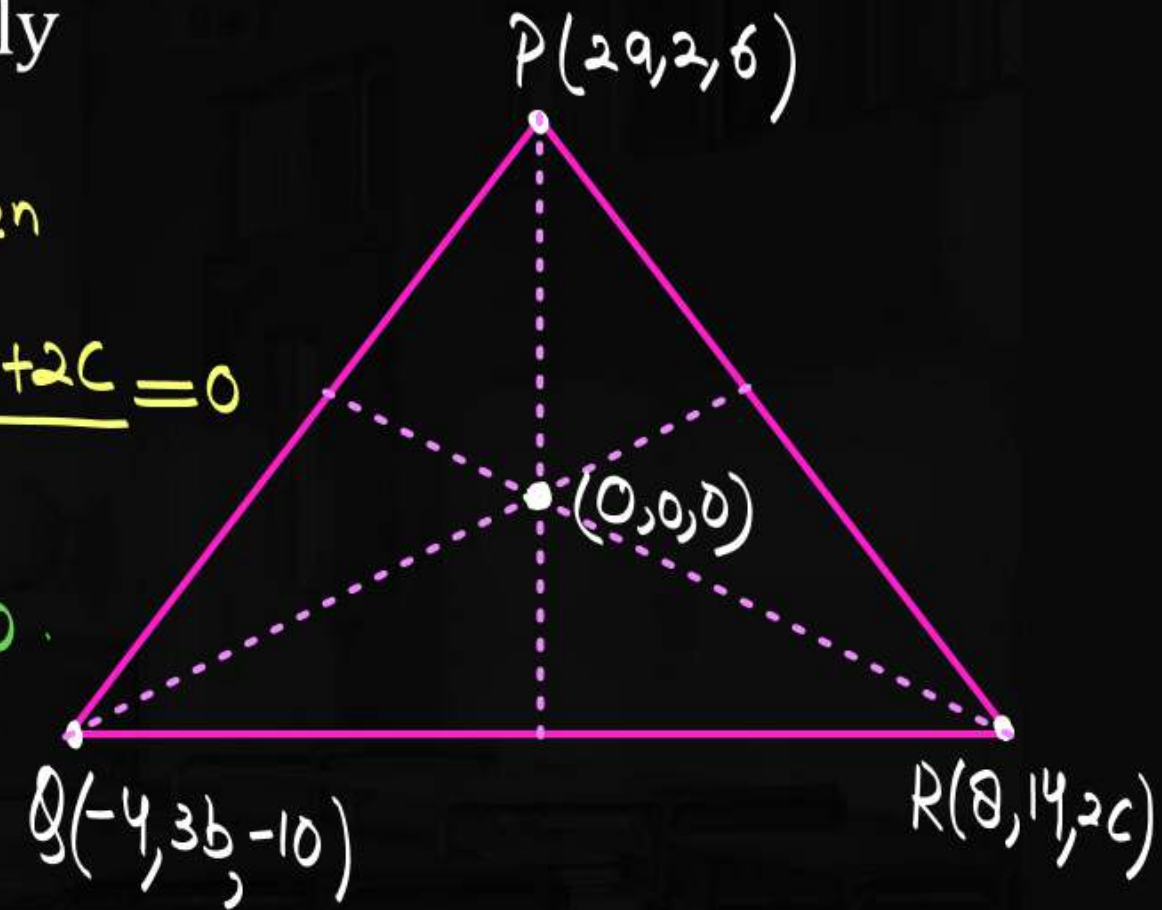
(Sol) Since $O(0,0,0)$ is the centroid of ΔPQR , then

$$\frac{2a - 4 + 8}{3} = 0, \quad \frac{2 + 3b + 14}{3} = 0, \quad \frac{6 - 10 + 2c}{3} = 0$$

$$\Rightarrow 2a + 4 = 0, \quad 16 + 3b = 0, \quad -4 + 2c = 0$$

$$\Rightarrow a = -2, \quad b = -\frac{16}{3}, \quad c = 2$$

$$\Rightarrow \boxed{a = -2, b = -\frac{16}{3}, c = 2}$$



EXERCISE – 11.2

QUESTION

Find the distance between the following pairs of points:

(i) $(2, 3, 5)$ and $(4, 3, 1)$

(ii) $(-3, 7, 2)$ and $(2, 4, -1)$

(iii) $(-1, 3, -4)$ and $(1, -3, 4)$

(iv) $(2, -1, 3)$ and $(-2, 1, 3)$.

H/W

QUESTION

Show that the points $(-2, 3, 5)$, $(1, 2, 3)$ and $(7, 0, -1)$ are collinear.

HP/W

QUESTION



Verify the following:

- (i) $(0, 7, -10)$, $(1, 6, -6)$ and $(4, 9, -6)$ are the vertices of an isosceles triangle.
- ✓ (ii) $(0, 7, 10)$, $(-1, 6, 6)$ and $(-4, 9, 6)$ are the vertices of a right angled triangle.
- ✓ (iii) $A(-1, 2, 1)$, $B(1, -2, 5)$, $C(4, -7, 8)$ and $D(2, -3, 4)$ are the vertices of a parallelogram.

(Sol) (i) let $A(0, 7, -10)$, $B(1, 6, -6)$ and $C(4, 9, -6)$ be the given points.

Then by Distance Formula,

$$AB = \sqrt{1+1+16} = 3\sqrt{2}$$

$$BC = \sqrt{9+9+0^2} = 3\sqrt{2}$$

$$AC = \sqrt{16+4+16} = 6$$

Here $AB + BC = 6\sqrt{2} > AC$

$$AB + AC = 3\sqrt{2} + 6 > BC$$

$$BC + AC = 3\sqrt{2} + 6 > AB$$

This shows that A, B, C are non-collinear, hence they form the vertices of some $\triangle ABC$. Note that

$AB = BC \Rightarrow \triangle ABC$ is an isosceles triangle

∴ A, B, C are the vertices of an isosceles triangle.

QUESTION



Find the equation of the set of points which are equidistant from the points $(1, 2, 3)$ and $(3, 2, -1)$.

(Sol) Suppose that $R(x, y, z)$ be the required point which is equidistant from $P(1, 2, 3)$ & $Q(3, 2, -1)$

$$\text{i.e. } PR = QR$$

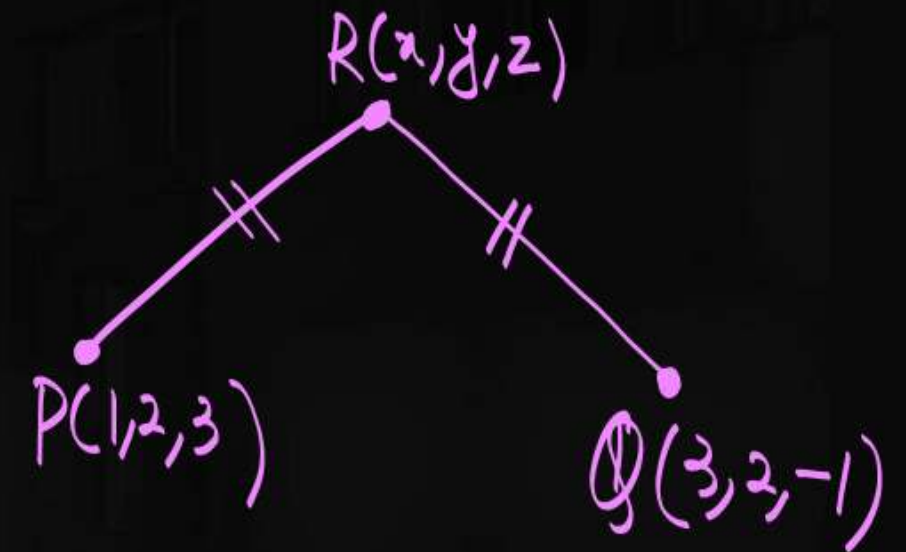
$$\Rightarrow PR^2 = QR^2$$

$$\Rightarrow (x-1)^2 + (y-2)^2 + (z-3)^2 = (x-3)^2 + (y-2)^2 + (z+1)^2$$

$$\Rightarrow x^2 - 2x + 1 + z^2 - 6z + 9 = x^2 - 6x + 9 + z^2 + 1 + 2z$$

$$\Rightarrow -2x + 6x = 2z + 6z$$

$$\Rightarrow 4x = 8z$$



$$\Rightarrow x = 2z$$

$$\therefore \text{Required set of points} = \{(x, y, z) : x = 2z\}$$

QUESTION

Find the equation of the set of points P , the sum of whose distances from $A(4, 0, 0)$ and $B(-4, 0, 0)$ is equal to 10.

H/W

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 11**

QUESTION

Three vertices of a parallelogram ABCD are $A(3, -1, 2)$, $B(1, 2, -4)$ and $C(-1, 1, 2)$. Find the coordinates of the fourth vertex.

(Sol) Suppose that $D(x, y, z)$ be the vertices of a parallelogram ABCD. Since we know that the diagonals of a parallelogram bisect each other, So O is the mid point of AC & BD.

$$\Rightarrow O\left(\frac{-1+3}{2}, \frac{-1+1}{2}, \frac{2+2}{2}\right) = O\left(\frac{x+1}{2}, \frac{y+2}{2}, \frac{z-4}{2}\right)$$

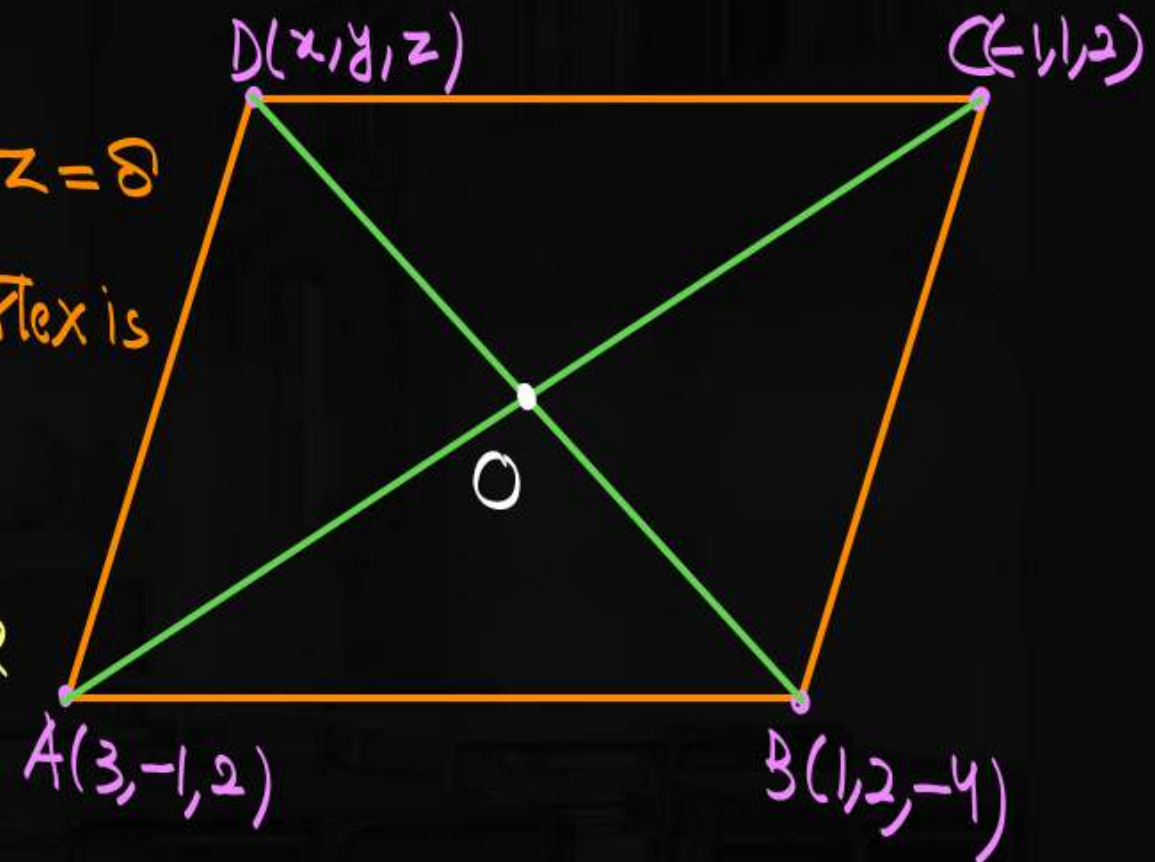
$$\Rightarrow O(1, 0, 2) = O\left(\frac{x+1}{2}, \frac{y+2}{2}, \frac{z-4}{2}\right)$$

$$\Rightarrow \frac{x+1}{2} = 1, \frac{y+2}{2} = 0, \frac{z-4}{2} = 2.$$

$$\Rightarrow x = 1, y = -2, z = 8$$

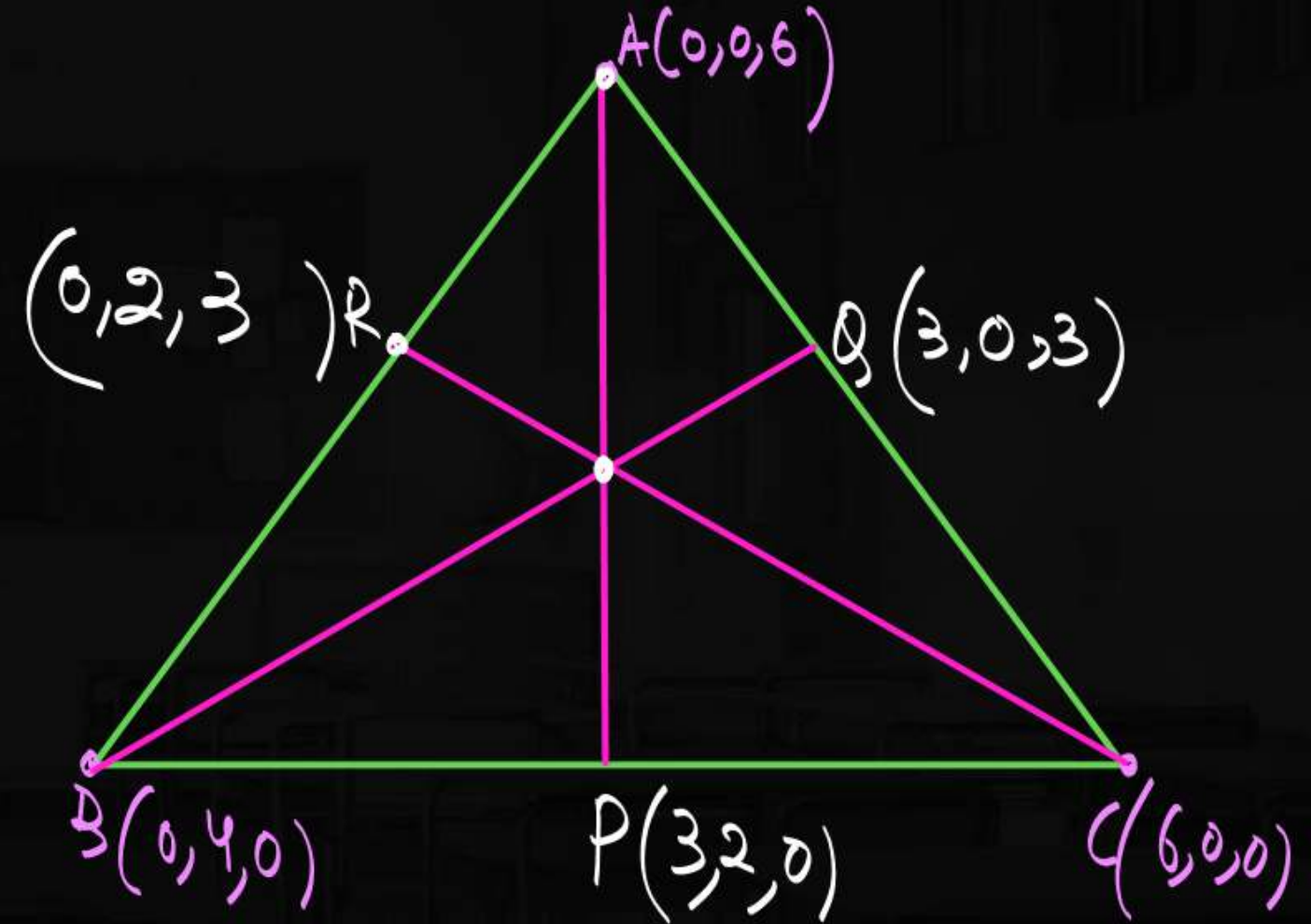
So the required vertex is $(1, -2, 8)$.

Hence the result



QUESTION

Find the lengths of the medians of the triangle with vertices $A(0, 0, 6)$ $B(0, 4, 0)$ and $C(6, 0, 0)$.



QUESTION

If the origin is the centroid of the triangle PQR with vertices $P(2a, 2, 6)$, $Q(-4, 3b, -10)$ and $R(8, 14, 2c)$ then find the values of a , b and c .

QUESTION

If A and B be the points $(3, 4, 5)$ and $(-1, 3, -7)$, respectively, find the equation of the set of points P such that $PA^2 + PB^2 = k^2$ where k is a constant.

(Sol) Suppose that $P(x, y, z)$ any point such that
 $PA^2 + PB^2 = k^2$.

$$\Rightarrow (x-3)^2 + (y-4)^2 + (z-5)^2 + (x+1)^2 + (y-3)^2 + (z+7)^2 = k^2$$

$$\Rightarrow \cancel{x^2} + \cancel{9} - \cancel{6x} + \cancel{y^2} + \cancel{16} - \cancel{8y} + \cancel{z^2} + \cancel{25} - \cancel{10z} + \cancel{x^2} + \cancel{2x} + \cancel{1} + \cancel{y^2} + \cancel{9} - \cancel{6y} + \cancel{z^2} + \cancel{14z} + \cancel{49} = k^2$$

$$\Rightarrow 2x^2 + 2y^2 + 2z^2 - 4x - 14y + 4z + 109 = k^2$$

Syed Humeera ★★★★★

Rasib Ayoub Rather ❤️❤️❤️❤️❤️

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Thank You



AD Nan

Zeeshan Misgar ★★★★★